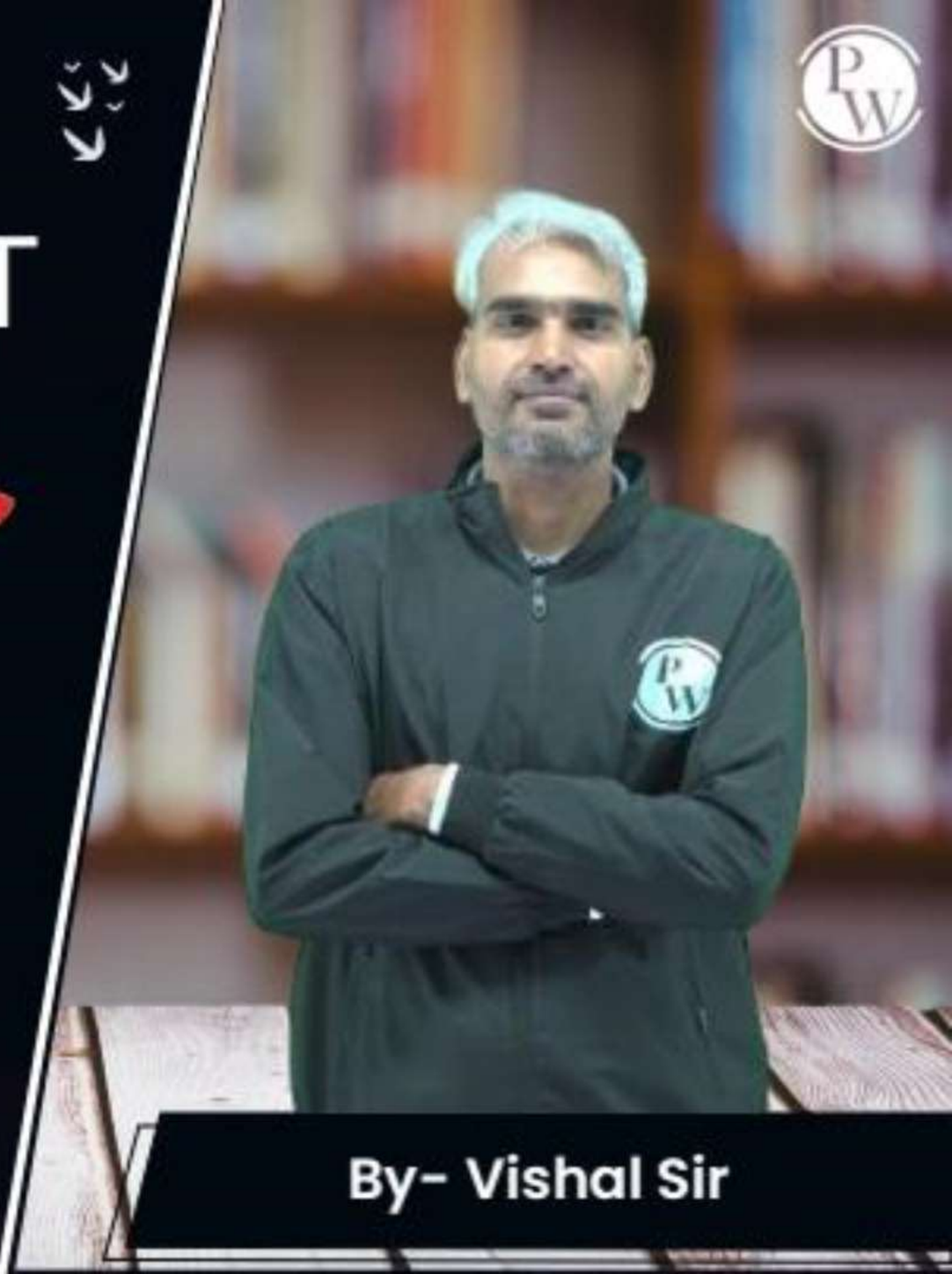
Computer Science & IT



Database Management System

Relational Model & Normal Forms

Lecture No. 11



By- Vishal Sir

Recap of Previous Lecture



- **Topic** Lossless Join decomposition
- **Topic** Projection, Selection, and Cross product
- **Topic** Natural join

$$R \bowtie S = \Pi_{\text{distinct Attributes}}\left(\sigma_{\text{Equality on Common Attributes}}(R \times S)\right)$$

- **Topic** Lossless Natural Join

Slide

Topics to be Covered



- Topic: Normal forms
- Topic: First normal form (1NF)
- Topic: Redundancy in relation because of FD
- Topic: Second normal form (2NF)
- Topic: Third normal form (3NF)
- Topic: Boyce codd normal form (BCNF)

Slide

#Q. Let R (A, B, C, D, E, F) be the relational schema with following FD set

F = { AB→C, BC→A, AC→B, AD→E, B→D, E→F)

Which of the following decomposition is lossless join decomposition.

(i){R1(ABC), R2(ABDE), R3(EF)}

(ii){R1(ABC), R2(ADF), R3(ACDE)}

(iii){R1(AB), R2(BC), R3(ABDE), R4(EF)}

$(R_1 \bowtie R_3) \bowtie R_2$ } ∴ Lossless Join decomposition

$R_1 \bowtie R_3$ → lossless

$\bowtie R_2$ → lossless

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lossless Join decomposition

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lossless Join decomposition

(iii){R1(AB), R2(BC), R3(ABDE), R4(EF)}

$R_1(ABC)$ $R_2(ABDE)$ $R_3(EF)$

$R_1 \cap R_2 = AB$

$(AB)^+ = \{A, B, C, D, E, F\}$

(AB) is S.K of both relations → lossless

$\therefore R_1$ & R_2 can be joined

$R_1 \bowtie R_2(ABCDE)$

$(R_1 \bowtie R_2) \cap R_3 = E \Rightarrow (E)^+ = \{E, F\}$

i.e E is a S.K of R_3 → lossless

$\therefore (R_1 \bowtie R_2)$ can be joined with R_3

$(R_1 \bowtie R_2) \bowtie R_3(ABCDEF)$

#Q. Let R (A, B, C, D, E, F) be the relational schema with following FD set

$F = \{ AB \rightarrow C, BC \rightarrow A, AC \rightarrow B, AD \rightarrow E, B \rightarrow D, E \rightarrow F\}$

$R_1(AB)$ $R_2(BC)$ $R_3(ABDE)$ $R_4(EF)$

$R_1 \cap R_2 = \{B\} \neq \{B,D\}$

Which of the following decomposition is lossless join decomposition.

Not a S.K. of R_1 or R_2 ∴ Can not be joined

(i){R1(ABC), R2(ABDE), R3(EF)}
lossless Join decomposition

(ii){R1(ABC), R2(ADF), R3(ACDE)}
lossless Join decomposition

(iii){R1(AB), R2(BC), R3(ABDE), R4(EF)}

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$R_1(AB)$ $R_2(BC)$ $R_3(ABDE)$ $R_4(EF)$

$R_1 \cap R_2 = (B)$ is a s.k of R_1 & R_2

$R_1 \bowtie R_3 (ABDE)$

$(R_1 \bowtie R_3) \cap R_2 = (B)^+ = \{B, D\}$

B is not a s.k

∴ Can not be joined

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(iii){R1(AB), R2(BC), R3(ABDE), R4(EF)}
lossy Join decomposition

$R_1(AB)$ $R_2(BC)$ $R_3(ABDE)$ $R_4(EF)$

$R_1 \cap R_2 = (AB)$ is a S.K of R_1 & R_2

$R_1 \bowtie R_3 (ABDE)$

$R_2 \cap (R_1 \bowtie R_3) = (B)^+ = (BD)$

Not a S.K

$R_2(BC)$ can not be joined in any order s.t. join is lossless.

$(R_1 \bowtie R_3) \bowtie R_4 (ABDEF)$

$(R_1 \bowtie R_3) \cap R_4 = (E)^+ = \{EF\}$

E is S.K of R_4 ∴ Join



Topic : Normalization

Normalization is the process of decomposing the relation into sub-relations s.t redundancy is reduced/eliminated

- Normal forms are used to describe the type of redundancy present in the relation.



Topic : Normal forms

There are various normal Forms

1 NF 2NF 3NF BCNF | 4NF

* Upto BCNF we try to eliminate the redundancies present in the relation because of Functional dependencies

* If the relation is in BCNF then, there will be 0% redundancy in the relation because of Functional dependencies, but a relation which is in BCNF may suffer from redundancies Present in the relation because of Multi-Valued dependencies

↑

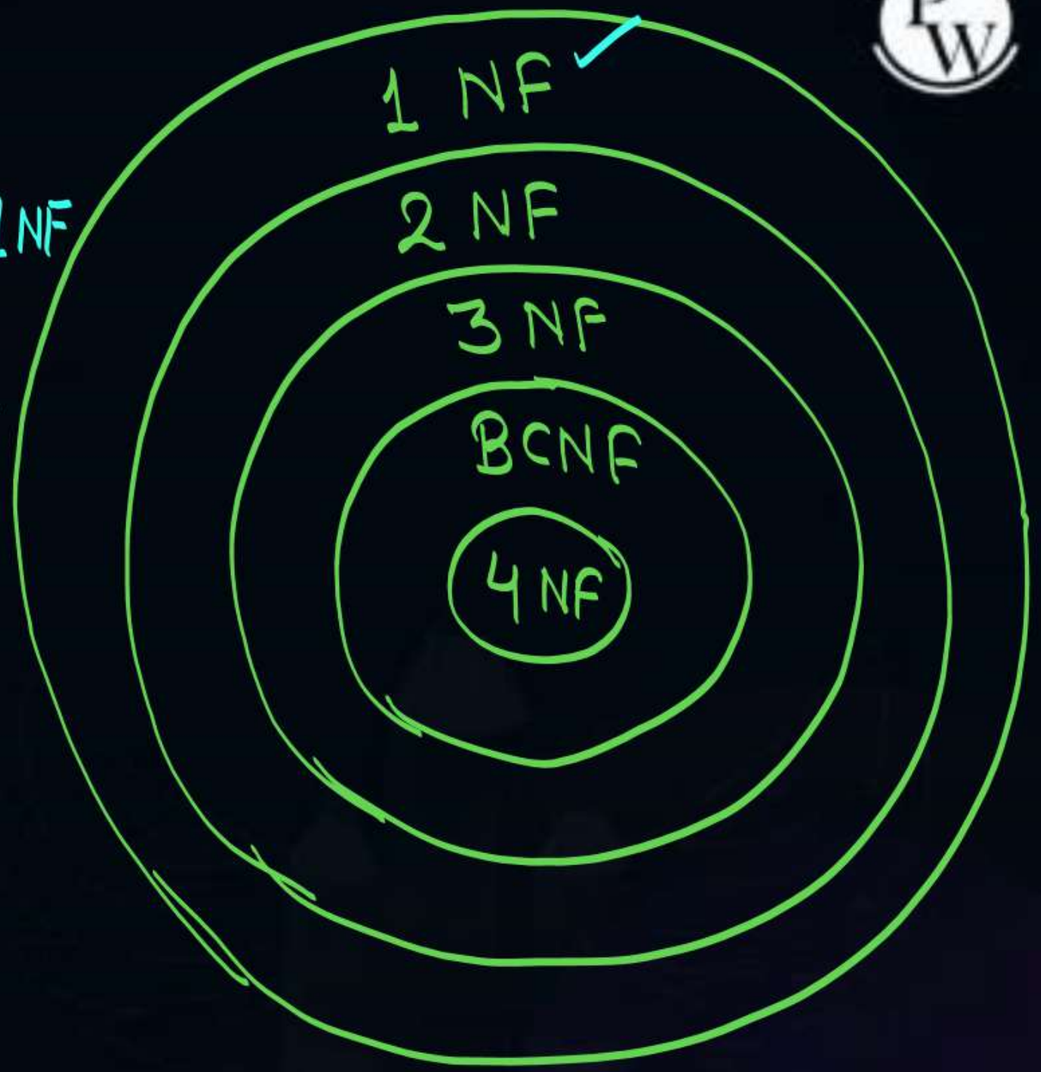
4NF is related Multi-valued dependencies

{ In 4F we will try to eliminate the redundancy Present in the relation because of Multi-Valued dependency (MVD)



Topic : Normal forms

- If relation is in 2NF then it is also in 1NF
- If relation is in 3NF then it is also in 2NF
- If relation is in BCNF then it is also in 3NF
- If relation is in 4NF then it is also in BCNF



Topic : First normal form (1NF)

A database is in 1NF only if it does not Contain any multi-valued attributes

Multi-valued attribute

Sid	Project
S_1	P_1, P_2
S_2	P_3, P_4
S_3	P_1

Multivalued attribute exists in the database
∴ Database is not in 1NF

Convert it into 1NF

i.e. Convert Multivalued attribute in Single-Valued attribute

Converted into single valued attribute

Sid	Project
S_1	P_1
S_1	P_2
S_2	P_3
S_2	P_4
S_3	P_1

No multivalued attribute exists., ∴ At least in 1NF



Topic : First normal form (1NF)

* In relational table no entry of the table can contain more than one value.

∴ By default normal form of relational database is 1NF.

- i.e. Every relation is at least in First normal form.



Topic : Redundancy in relation because of FD

Rule 1 :- A non-trivial functional dependency '$X \rightarrow Y$' does not cause any redundancy in the relation if 'X' is a Super key of the relation.

X → it is a SK {∵ Values will be unique}

Y → A B C

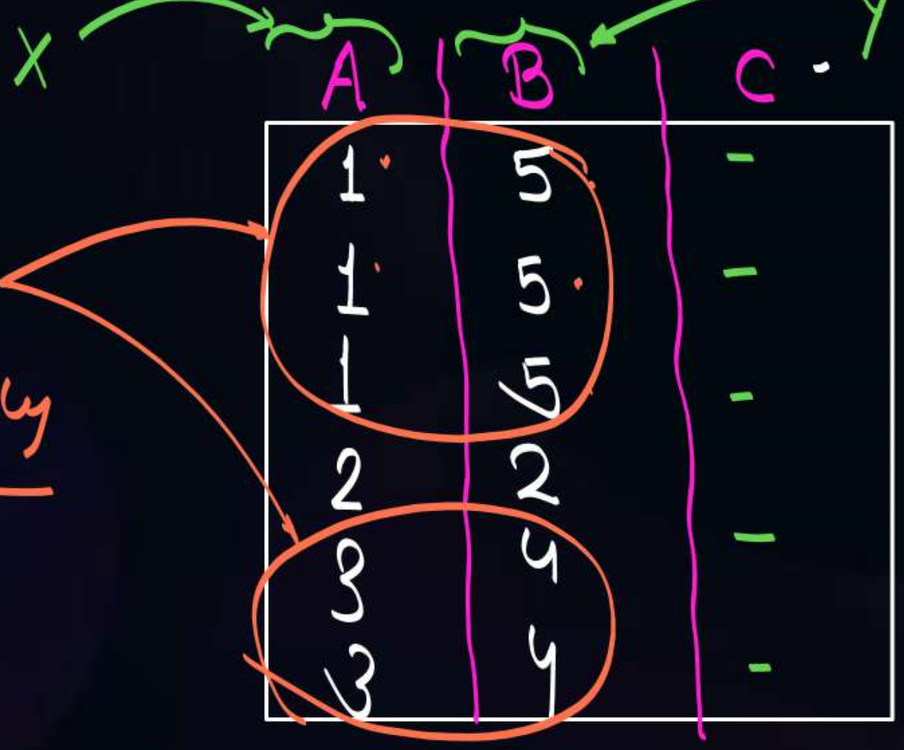
A	B	C
1	2	1
3	1	1
2	2	1
3	2	1
4	1	2

it is not redundancy
it is information



Topic : Redundancy in relation because of FD

Rule 2 :- A non-trivial functional dependency 'X → Y' will always cause redundancy in the relation if 'X' is not a Super key



Note: ① In FD $X \longrightarrow Y$,

if X is not a Super key, then Y can never be a superkey.

② Neither X can be a S.K {or C.K} nor Y can be a S.K {or C.K}

if $X \longrightarrow Y$ causes redundancy



Topic : Redundancy in relation because of FD

Possible types of functional dependencies which may cause redundancy in the relation

$X \rightarrow Y$

① Proper subset of C.K → Non-prime attribute

② Non-prime attribute → Non prime attribute

③ Proper subset of one C.K → Proper subset of another C.K.



Topic : Redundancy in relation because of FD

Possible types of functional dependencies which may cause redundancy in the relation

$X \rightarrow Y$

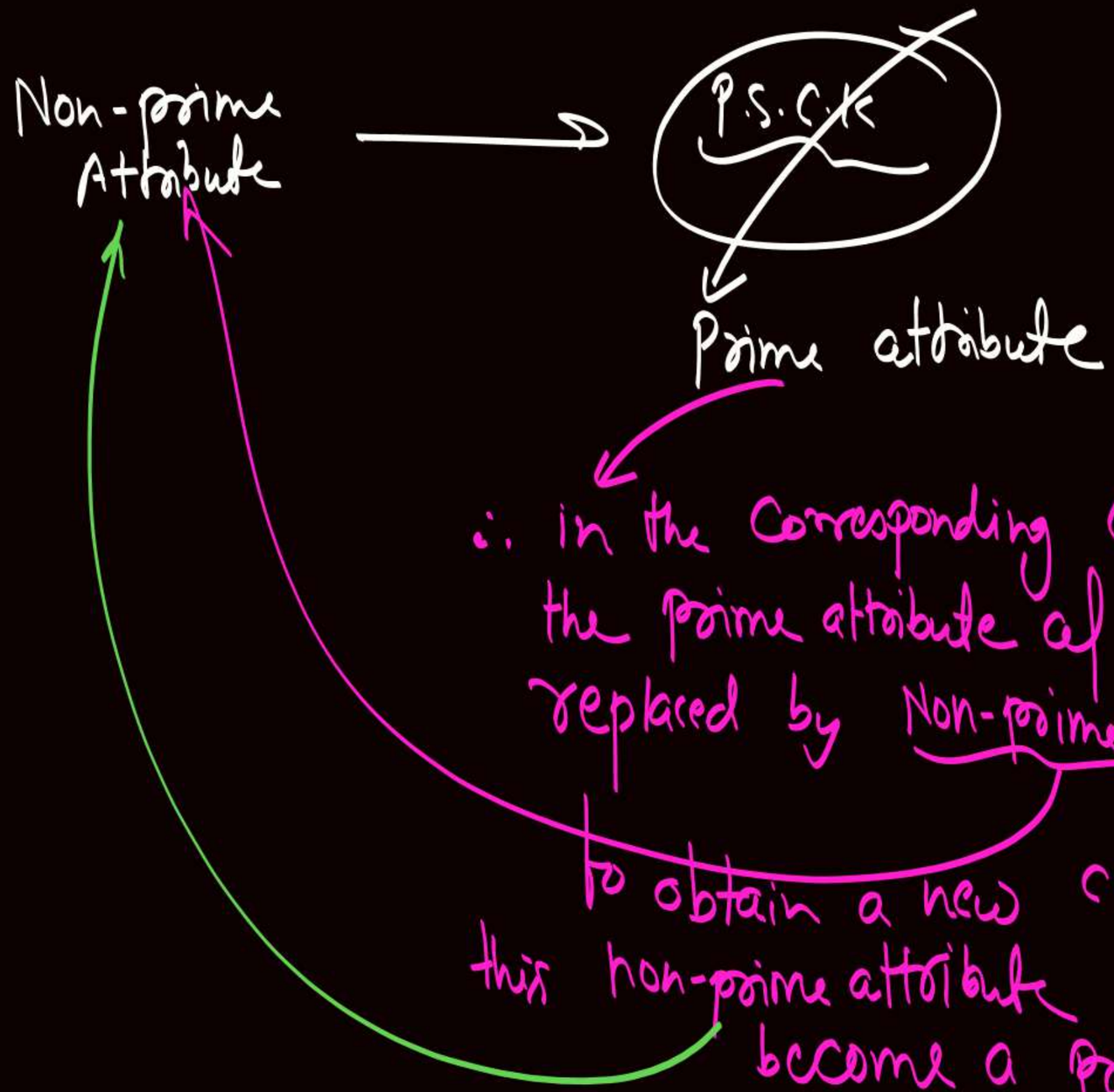
① Proper subset of C.K → Non-prime attribute

② Non-prime attribute → Non prime attribute

③ Proper subset of one C.K → Proper subset of another C.K.

it is never Possible

④ Non-prime attribute → Proper subset of C.K.


Non-prime Attribute
P.S.C.K
Prime attribute
∴ in the Corresponding C.K.
the prime attribute of can be
replaced by Non-prime attribute of L.H.S.
to obtain a new C.K., and
this non-prime attribute will also
become a prime attribute



Topic : Redundancy in relation because of FD

Possible types of functional dependencies which may cause redundancy in the relation

$X \rightarrow Y$

① Proper subset of C.K → Non-prime attribute

② Non-prime attribute → Non prime attribute

③ Proper subset of one C.K → Proper subset of another C.K.

④

Such FD is not possible

⑤ Proper subset of a C.K → Proper subset of same C.K.

- Let (AB) is a candidate key and

$A \longrightarrow B$ is a FD in the FD set

A: Proper subset of C.K

B: Proper subset of same C.K

Whatever that can be determined using attribute B can also be determined using attribute A.

∴ (AB) is not minimal because 'B' can actually be removed from the set, and hence (AB) can not be the C.K because it is not minimal



Topic : Redundancy in relation because of FD

Possible types of functional dependencies which may cause redundancy in the relation



⑥ Proper subset of C.K → (Proper subset of another C.K + Non-prime attribute)

We can split it into two parts

P.S. C.K → P.S. of another C.K (it is our Case ③)

&

P.S. C.K → Non prime attribute (it is our Case ①)



Topic : Redundancy in relation because of FD

Possible types of functional dependencies which may cause redundancy in the relation

$X \longrightarrow Y$

(7) (Proper Subset of a c.k + Non-prime attribut) $\longrightarrow$ Any thing may be present in R.HS.

We will focus on this type of FDs during the discussion of 3 NF.



Topic : Redundancy in relation because of FD

Possible types of functional dependencies which may cause redundancy in the relation

$X \rightarrow Y$

IMP

1. Proper subset of C.K → Non-prime attribute
2. Non-prime attribute → Non prime attribute
3. Proper subset of one C.K → Proper subset of another C.K.

We only need to remember these three types of FDs



2 mins Summary

- **Topic** Normal forms
- **Topic** First normal form (1NF)
- **Topic** Redundancy in relation because of FD
- **Topic** Second normal form (2NF)
- **Topic** Third normal form (3NF)
- **Topic** Boyce codd normal form (BCNF)

THANK - YOU